Algebra 1 — Probability Practice (20 Problems)

Show your reasoning. Exact fractions are fine; decimals optional.

Quick Review

Outcome: one possible result of an experiment. **Sample space:** set of all outcomes.
Event: a set of outcomes we care about. **Simple probability:** $P(E) = \dfrac{\#\text{favorable}}{\#\text{total}}$.
Independent: one event does not change the probability of the other (e.g., coin vs. die).
Dependent: one changes the other (e.g., drawing without replacement).
Multiplication rule (independent): $P(A \text{ and } B) = P(A) \cdot P(B)$.
Addition rule (disjoint): $P(A \text{ or } B) = P(A) + P(B)$ if $A \cap B = \varnothing$.
General addition: $P(A \text{ or } B) = P(A) + P(B) - P(A \cap B)$.
Permutations: order matters ${}_nP_r = \dfrac{n!}{(n-r)!}$. **Combinations:** order doesn't ${}_nC_r = \dfrac{n!}{r!(n-r)!}$.
Complement: $P(\text{not } A) = 1 - P(A)$.

1. **Sample space.** Flip two fair coins. List the sample space S and the event $E =$ "exactly one head." Then find $P(E)$.

2. **Simple probability.** A number cube (1–6) is rolled. Find $P(\text{roll} > 3)$.

3. **Bag of marbles.** A bag has 5 red, 3 blue, and 2 green marbles. One is drawn at random. Find $P(\text{green})$ and $P(\text{not blue})$.

4. **Complement check.** In Problem 2, use a complement to recompute $P(\text{roll} > 3)$ and verify it matches.

5. **Independent?** You flip a fair coin and roll a fair die. Are the events $A =$"Heads" and $B =$"Even" independent? Justify and find $P(A \text{ and } B)$.

6. **Without replacement (dependent).** From the marble bag in Problem 3, two marbles are drawn *without replacement*. Find $P(\text{both red})$.

7. **With replacement (independent).** Same bag, but now you draw a marble, replace it, mix, and draw again. Find $P(\text{red then red})$.

8. **Two dice sum.** Two fair dice are rolled. Find $P(\text{sum} = 7)$ and $P(\text{sum} \geq 10)$.

9. **At least one.** Three fair coin flips. Find $P(\text{at least one head})$ using a complement.

10. **Survey (overlap).** In a class, $P(\text{likes pizza}) = 0.70$, $P(\text{likes tacos}) = 0.55$, and $P(\text{both}) = 0.35$. Find $P(\text{pizza or tacos})$ and $P(\text{neither})$.

11. **Permutations.** How many different 5-letter "arrangements" can be made from distinct letters A, B, C, D, E? How many start with A?

12. **Permutations: no repeats.** Form a 4-digit code using digits 0–9 with no repeated digits, and the first digit cannot be 0. How many codes?

13. **Combinations (choices).** A sandwich shop offers 8 toppings. How many different 3-topping sandwiches can you make (order doesn't matter)?

14. **Combo with restriction.** From 10 students, choose a 3-person committee, but Alex must be on the committee. How many committees?

15. **Independent vs dependent (identify).** For each scenario, say "independent" or "dependent" and give a sentence why.
(a) Draw one card from a deck, *replace*, draw again.
(b) Draw one card, *keep it out*, draw again.
(c) Spin a spinner and roll a die.

16. **Tree reasoning.** A spinner has 3 equal sections: R, G, B. You spin twice.
(a) How many outcomes in the sample space?
(b) Find $P(\text{at least one } R)$.
(c) Find $P(\text{both spins different colors})$.

17. **Union (disjoint).** A jar has 6 yellow and 4 purple candies. One candy is drawn. What is $P(\text{yellow or purple})$ and why is it so simple here?

18. **From counts to probability.** In 200 survey responses, 120 chose tea, 90 chose coffee, and 40 chose both. If one response is picked at random, what is $P(\text{tea or coffee})$?

19. **Mixed counting.** A password has 2 letters (A–Z) followed by 3 digits (0–9).
(a) How many if letters and digits can repeat?
(b) How many if no repeats at all (letters cannot repeat among themselves; digits cannot repeat among themselves; and no letter–digit overlap rule needed)?

20. **Application.** A box contains 4 defective and 6 good bulbs. Two bulbs are drawn *without replacement.*
(a) P(both good)   (b) P(at least one defective).